

ETIOLOGY

Most cases of erythema multiforme (EM) are related to infections. Herpes simplex virus (HSV) is definitively the most common cause, particularly in recurrent cases. The causal relationship between HSV and EM is well established through clinical experience, epidemiology, detection of HSV DNA in EM lesions, and the prevention of EM with antiviral suppression of HSV recurrences. A clinical link with herpes can be identified in approximately half of EM cases. Additionally, 10% to 40% of cases without apparent herpes symptoms are also found to be herpes-related, as HSV DNA is detectable in EM lesions by polymerase chain reaction (PCR) testing. EM eruptions typically begin around 7 days after a herpes recurrence, although the interval can be shorter. Not every symptomatic HSV recurrence leads to EM, and EM can even follow asymptomatic HSV reactivation, making the connection easy to overlook by both patients and clinicians. HSV-1 is the primary subtype involved, though HSV-2 can also trigger EM; the relative frequency reflects general HSV subtype prevalence in the population.

Mycoplasma pneumoniae is the second major cause of EM and may be the leading cause in children. EM associated with *M. pneumoniae* often presents with atypical and more severe features compared to HSV-related cases. Establishing the link can be challenging, as clinical and radiological signs of pneumonia may be mild, and direct detection of *M. pneumoniae* is uncommon. PCR testing of throat swabs offers the highest sensitivity. Serologically, the diagnosis is supported by the presence of immunoglobulin M (IgM) antibodies or a greater than twofold rise in immunoglobulin G (IgG) antibodies to *M. pneumoniae*. EM related to *M. pneumoniae* can recur.

Other infectious agents have been occasionally linked to EM in isolated cases or small series, but evidence for causality remains circumstantial. These include orf virus, varicella-zoster virus, parvovirus B19, hepatitis B and C viruses, and infectious mononucleosis, as well as various bacterial and viral infections.

Immunizations have also been reported as potential triggers, particularly in children.

Drug-induced EM with mucous membrane involvement is rare. It remains uncertain whether such eruptions represent true EM or mimics such as annular urticaria or maculopapular drug eruptions with target-like lesions.

Idiopathic EM refers to cases where no HSV infection or other identifiable cause is found. These are relatively common in routine practice. However, HSV DNA has been detected by PCR in up to 40% of recurrent "idiopathic" cases. Some of these respond to prophylactic antiviral therapy, suggesting asymptomatic HSV reactivation as the trigger, while others do not, indicating possible alternative mechanisms.

EM-like dermatitis can also arise from contact sensitization. Despite clinical and histopathological similarities, these eruptions should be considered mimics of EM rather than true EM.

PATHOGENESIS

The pathogenesis of EM has been most thoroughly studied in herpes-associated cases. It is unclear whether the same mechanisms apply to EM caused by other agents.

Infectious HSV has never been isolated from EM lesions. However, HSV DNA fragments—consistently including the viral polymerase (Pol) gene—have been detected in lesional keratinocytes using PCR. These fragments are localized primarily in basal and lower spinous layer keratinocytes, where the HSV Pol protein is also expressed. This leads to recruitment of HSV-specific T cells, including cytotoxic T cells. The initial virus-specific immune response is followed by a broader, non-specific inflammatory reaction mediated by autoreactive T cells.